

ECON 101—Introduction to Microeconomics
Fall 2019

Midterm I

September 23, 2019

VERSION I

Name:

NU ID number:

Instructions:

- Answer all questions.
- Fill in the answers on the answer sheet.

Part I: Multiple Choice

Please record all questions in the answer sheet. Each question is worth 2 points for a total of 40 points.

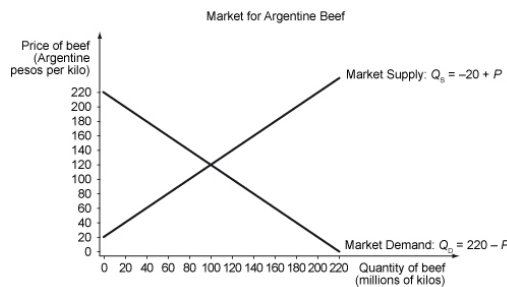
1. The total cost of a firm is \$50, the average variable cost is \$2, and the average fixed cost is \$3. How many units of output does the firm produce?
 - (a) 15 units
 - (b) 5 units
 - (c) 18 units
 - (d) 10 units

2. A perfectly competitive firm initially is earning zero economic profit. Then, a decrease in demand for the firm's product occurs. Of the following, in the long run which action listed below is the firm *most* likely to take?
 - (a) Increase the size of its plant.
 - (b) Increase the quantity it produces.
 - (c) Exit the market.
 - (d) Increase its advertising to increase the demand for its product.

Quantity (tattoos per hour)	Total Cost (dollars per hour)
0	10
1	25
2	35
3	50
4	70
5	95
6	125

3. Archibald's Tattoos is a perfectly competitive firm. The firm's costs are shown in the table above. If the market price of a tattoo is \$17.50, what is the firm's economic profit?
- \$2.5 per hour
 - \$12.50 per hour
 - \$10 per hour
 - zero
4. Initially, a perfectly competitive industry that has 1,000 firms is in long-run equilibrium. Then 100 firms in the industry adopt a new technology that reduces the average cost of producing the good. In the short run, the price _____, firms with the new technology make _____ economic profit, and firms with the old technology _____.
- remains the same; zero; incur economic losses
 - falls; positive; incur economic losses
 - remains the same; positive; make normal profit
 - remains the same; positive; incur economic losses

Scenario: The retail market for beef in Argentina is given by the following demand and supply schedules, where the price is in Argentine pesos per kilo and the quantity is millions of kilos per year. Demand $Q_D = 220 - P$, Supply $Q_S = -20 + P$.



With the previous information please answer questions 5 and 6

5. Refer to the scenario above. The market equilibrium quantity is _____ million kilos of beef per year and the market equilibrium price is _____ pesos per kilo.
- 60; 80
 - 100; 120
 - 80; 140
 - None of the above
6. Refer to the scenario above. At the competitive market equilibrium, consumer surplus is _____ million pesos, producer surplus is _____ million pesos, and social surplus is _____ million pesos.
- 5,000; 10,000; 15,000
 - 10,000; 5,000; 15,000
 - 5,000; 5,000; 10,000
 - 10,000; 10,000; 20,000

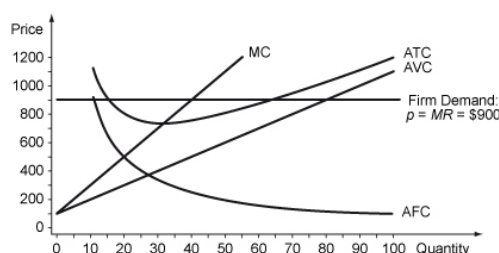
7. Wendy spends \$30 a week on movies and magazines. The price of a movie is \$8, the price of a magazine is \$2, and Wendy sees 3 movies a week and buys 3 magazines. The price of a magazine now increases to \$4 and Wendy's brother gives her \$6 a week so that she can still see 3 movies a week and buy 3 magazines. In this situation, Wendy will see _____ movies and buy _____ magazines.
- 3; 3
 - less than 3; more than 3
 - more than 3; fewer than 3
 - less than 3; less than 3

Scenario: A perfectly competitive firm has the following cost functions:

Total Cost: $TC(q) = 10,000 + 100q + 10q^2$,

Marginal Cost: $MC(q) = 100 + 20q$,

and it faces a market price of \$900 per unit. The average and marginal cost curves, in addition to the marginal revenue curve, are shown in the following figure.

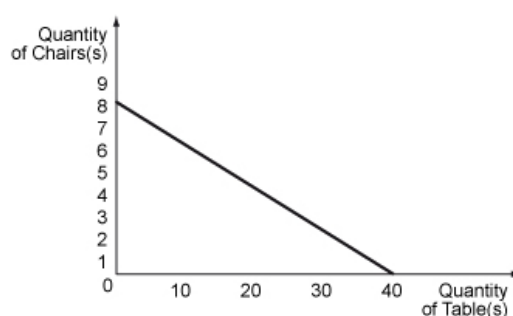


With the previous information please answer questions 8–10.

8. Refer to the scenario above. The firm's average total cost is given by _____.
- $100 + 20q$
 - 10,000
 - $\frac{10,000}{q} + 100 + 10q$
 - $100 + 10q$
9. The firm maximizes profits by _____.
- producing 40 units of output
 - producing 80 units of output
 - shutting down
 - producing 11 units of output
10. At the profit-maximizing level of output, producer surplus is _____ and the firm's profits are _____.
- \$16,000; \$6,000
 - \$32,000; \$16,000
 - \$0; \$0
 - None of the above
11. Suppose Z is a normal good. The equilibrium price and quantity of Z in the year 2013 was \$25 and 60 units, respectively. In 2017, the equilibrium price of Z had increased to \$35 and the equilibrium quantity had increased to 70 units. Other things remaining the same, which of the following could explain this change?
- A leftward shift of the demand curve for Z .
 - A leftward shift of the supply curve of Z .
 - A rightward shift of the supply curve of Z .
 - A rightward shift of the demand curve for Z .

12. Consider a market where the demand curve is downward sloping and the supply curve is upward sloping (so they are neither vertical nor horizontal). If the consumers' willingness to pay for the hundredth unit and the seller's willingness to accept for the 175th unit are both \$5.00, then _____.
- (a) there is an excess supply of 75 units at the price of \$5.00
 - (b) the equilibrium price is \$5.00
 - (c) there is an excess demand of 75 units at the price of \$5.00
 - (d) the equilibrium quantity is 100 units

The following figure displays John's budget constraint when he spends his income on tables and chairs.



13. Refer to the figure above. Which of the following statements is true?
- (a) The opportunity cost of buying one chair is 5 tables.
 - (b) The opportunity cost of buying one table is 7 chairs.
 - (c) The opportunity cost of buying tables increases as more tables are bought.
 - (d) The price of tables is more than the price of chairs.

14. Gasoline is a normal good. If the price of gasoline falls, a consumer buys more gasoline because of

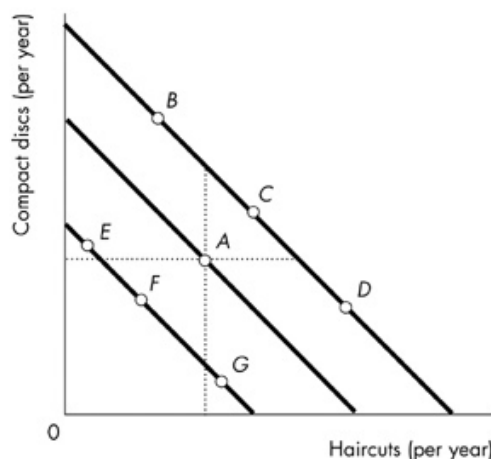
- (a) only an income effect.
- (b) only a substitution effect.
- (c) an increase in the marginal rate of substitution.
- (d) both an income effect and a substitution effect.

The following table shows the marginal benefit that Marcus derives by consuming different quantities of hotdog and soda. The price of a hotdog is \$3, and the price of a soda is \$1.

Quantity	Soda	Hotdog
	Marginal Benefit (\$)	Marginal Benefit (\$)
1	10	18
2	8	12
3	6	6
4	4	3
5	2	1
6	1	0.6

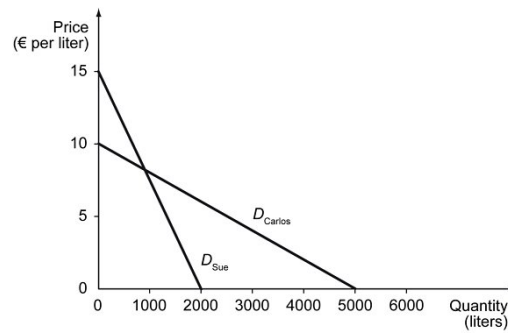
15. Refer to the table above. What is the maximum amount of consumer surplus that Marcus can get, if he has \$14 budget to spend on these two goods?
- (a) \$40
 - (b) \$14
 - (c) \$52
 - (d) \$8

16. Ryan buys 4 pairs of jeans when the price is \$25. He buys 3 pairs when the price increases to \$50. Ryan's percentage change in quantity demanded is _____, when the price increase by _____ percent, and thus his price elasticity of demand is _____.
- (a) +25%; -100%; -4
 - (b) -25%; +100%; -0.25
 - (c) -15%; +75%; -0.25
 - (d) -25%; +100%; -4



17. In the above figure, as Brendan's income falls his consumption bundle moves from point A to point E. This implies that for Brendan, compact discs are _____ good and haircuts are _____ good.
- (a) an inferior; an inferior
 - (b) an inferior; a normal
 - (c) a normal; a normal
 - (d) a normal; an inferior
18. Which of the following is most likely to be an example of causation?
- (a) A firm producing compact fluorescent light bulbs installs new machinery. Their per-day production of light bulbs increases.
 - (b) A soccer player scores four goals when he wears red socks. He concludes that the red socks helped him score the goals.
 - (c) A student wins money by scratching a ticket with a particular coin. He decides to scratch all tickets with the same coin in the future.
 - (d) The crime rate is high in a country. The literacy rate is high as well.
19. A firm is seeing a \$500 loss in the short run. The fixed cost of operation for this firm is \$1,000. What is the best decision for this firm in the short run?
- (a) This firm should shut down production.
 - (b) This firm should not shut down production.
 - (c) This firm should produce more than what it is currently producing.
 - (d) There is not enough information provided in the answer.

The figure below shows Sue's and Carlos' demand curves for gasoline.



20. Refer to the figure above. The total demand curve has the vertical intercept at _____ and the horizontal intercept at _____, and its slope _____.
- (a) 15.00 euros; 7,000 liters; changes at 10.00 euros.
 - (b) 15.00 euros; 7,000 liters; changes at 8.12 euros.
 - (c) 18.00 euros; 7,000 liters; is twice as steep as Sue's demand curve
 - (d) 10.00 euros; 7,000 liters; is half as steep as Carlos's demand curve

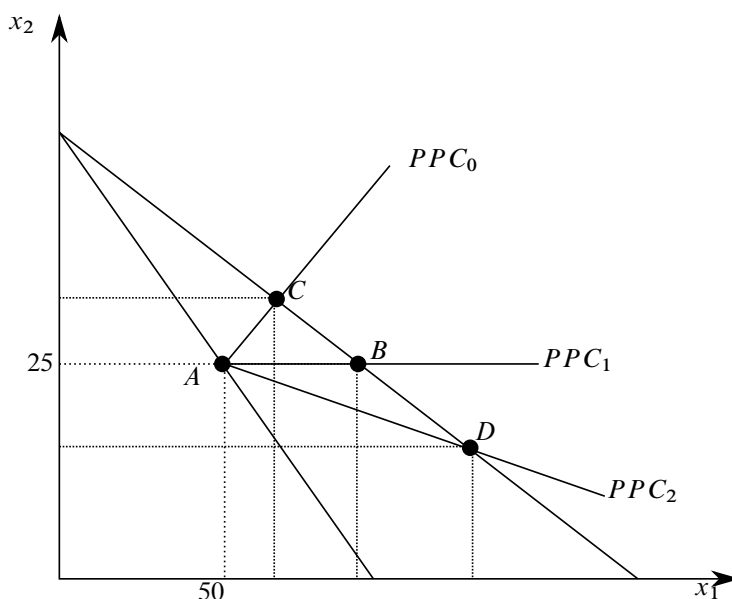
End of Multiple Choice Section: Please don't forget to record your answers in the answer sheet!

Continue to Part 2

Part II: Open problems

Instructions: Write your answers in the space provided in the **answer sheet**.

1. Consider the following scenario as depicted by the following figure.



There are three consumers, denoted by $i = 0, 1, 2$ which can consume two goods, x_1 and x_2 . The consumers have the same Income of \$100 dollars. The price of x_1 is \$1 dollar and the price of x_2 is \$2. Each consumer initially chooses the initial consumption bundle $x_1 = 50, x_2 = 25$ which is denoted by point A . Suppose that there is a reduction in the price of x_1 to \$0.5. After this price decrease, the optimal consumption bundle of consumer 0 is at point C , the optimal bundle of consumer 1 is at point B and the optimal bundle of consumer 2 is at point D . In the graph you can see the depiction of their *price consumption curves*, denoted by PPC_i for $i = 0, 1, 2$.

Suppose that a market researcher tells you that after estimating the own price elasticities (arc) for the demand of good x_1 for each consumer, he has found the following:

- $|\epsilon_{x_1}^{P_1}| = 1.5$
- $|\epsilon_{x_1}^{P_1}| = 1$
- $|\epsilon_{x_1}^{P_1}| = 0.6$

Unfortunately, the market researcher has been very careless and he forgot to record which elasticity (arc) corresponds to each consumer.

- (a) (5 points) With the data above, match each elasticity with its corresponding consumer and explain why?
 - (b) (10 points) What does the own price elasticity and the slope of the PPC curves tell us about the cross-price elasticity of good x_2 with respect to p_1 ?
 - (c) (15 points) Using the previous information, compute the optimal consumption bundles C , B and D of each consumer.
2. Suppose that an economy has two individuals, Alejandro and Oleg. Alejandro's demand curve is given by $q_D^A = 100,000 - p$ and Oleg's demand curve is given by $q_D^O = 20,000 - 0.5P$. Furthermore, there are initially 1,000 identical firms with the following cost structure:

Firm's Output	1	2	3	4	5
MC	11	11.13	12.00	13.63	16.00
ATC	13.50	12.25	12.00	12.19	12.70
AVC	11.25	11.13	11.25	11.63	12.25

- (a) (6 points) Perform horizontal summation of these demand curves to find the market demand curve. Denote this curve by Q_D

- (b) (6 points) What is the firm's shutdown point and what is the firm's supply schedule?
- (c) (6 points) What is the market price and quantity?
- (d) (6 points) Will firms enter or exit?
- (e) (6 points) Calculate the long-run equilibrium price, quantity, and number of firms.